
```

%Testing for part 3. This would be easier as a function
%Alas, I am lazy and so I do more work as a consequence

rect_pts = [1 2 2 1 1;
            2 2 5 5 2;
            3 3 3 3 3];

plane1 = plane_rotation_mat(1,pi/4);
plane2 = plane_rotation_mat(2,pi/4);
plane3 = plane_rotation_mat(3,pi/4);
hom_rect_pts = r3_to_hom(rect_pts);
plane1_rec = plane1 * hom_rect_pts;
plane2_rec = plane2 * hom_rect_pts;
plane3_rec = plane3 * hom_rect_pts;
final_rec1 = hom_to_r3(plane1_rec);
final_rec2 = hom_to_r3(plane2_rec);
final_rec3 = hom_to_r3(plane3_rec);

plot3(rect_pts(1,:),rect_pts(2,:),rect_pts(3,:), ...
      final_rec1(1,:),final_rec1(2,:),final_rec1(3,:), ...
      final_rec2(1,:),final_rec2(2,:),final_rec2(3,:), ...
      final_rec3(1,:),final_rec3(2,:),final_rec3(3,:))

axis equal

function rotate = plane_rotation_mat(iplane,gamma)
    if iplane == 1
        rotate = [cos(gamma) -sin(gamma) 0 0;
                  sin(gamma) cos(gamma) 0 0;
                  0 0 1 0;
                  0 0 0 1]
    end
    if iplane == 2
        rotate = [cos(gamma) 0 -sin(gamma) 0;
                  0 1 0 0;
                  sin(gamma) 0 cos(gamma) 0;
                  0 0 0 1]
    end
    if iplane == 3
        rotate = [1 0 0 0;
                  0 cos(gamma) -sin(gamma) 0 ;
                  0 sin(gamma) cos(gamma) 0;
                  0 0 0 1]
    end
end

function r4 = r3_to_hom(X)
    e = size(X);
    r4 = [X;ones(1,e(1,2))];
end

function r3 = hom_to_r3(X)
    e = size(X);

```

```
    r3 = eye(3,4) * X;  
end
```

```
rotate =
```

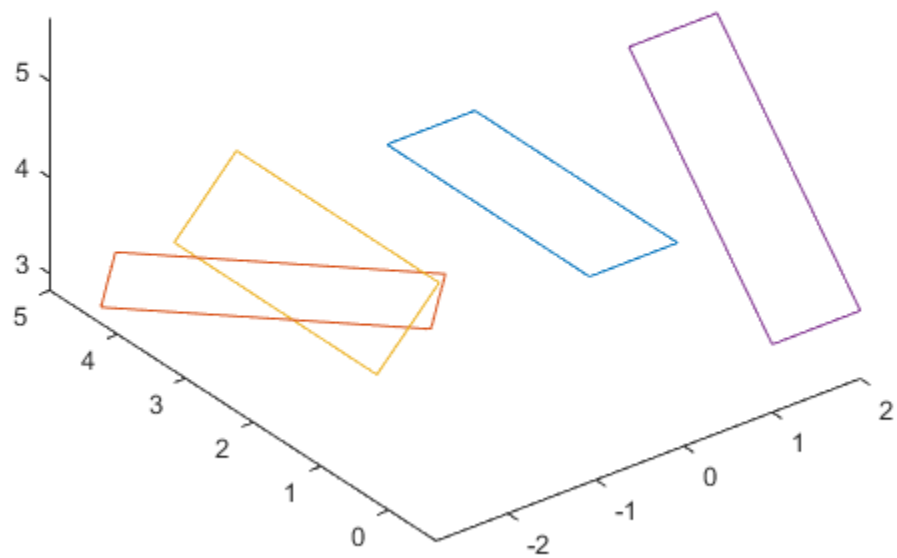
```
    0.7071    -0.7071         0         0  
    0.7071     0.7071         0         0  
         0         0     1.0000         0  
         0         0         0     1.0000
```

```
rotate =
```

```
    0.7071         0    -0.7071         0  
         0     1.0000         0         0  
    0.7071         0     0.7071         0  
         0         0         0     1.0000
```

```
rotate =
```

```
    1.0000         0         0         0  
         0     0.7071    -0.7071         0  
         0     0.7071     0.7071         0  
         0         0         0     1.0000
```



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